

Multiple matrix transforms

Basic skills

Q1. a) $\begin{pmatrix} 3 & -1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} -2 & 2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -7 & 6 \\ -4 & 6 \end{pmatrix}$

b) $\begin{pmatrix} -1 & 4 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix} = \begin{pmatrix} 10 & 21 \\ 6 & 9 \end{pmatrix}$

c) $\begin{pmatrix} 2 & 2 & 6 \\ 1 & -1 & 4 \\ 0 & 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -5 \\ 0 & 2 & 3 \\ -4 & 1 & 4 \end{pmatrix} = \begin{pmatrix} -22 & 10 & 20 \\ -15 & 2 & 8 \\ 4 & 3 & 2 \end{pmatrix}$

Q2. a) Stretch y by 4 so $\begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$

b) rot clockwise $30^\circ =$ Anticlockwise 330°
so by formula $\begin{pmatrix} \cos 330^\circ & -\sin 330^\circ \\ \sin 330^\circ & \cos 330^\circ \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$
 $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

Q3. a)  $\neq$ negative so $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

b) by formula

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 30^\circ & -\sin 30^\circ \\ 0 & \sin 30^\circ & \cos 30^\circ \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

Competency skills

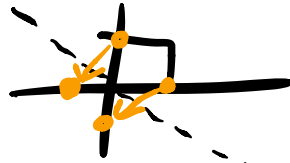
Q4. $S = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$ stretches x by 4

$T = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$ stretch y by 3

As want S then T we pre multiply S by T

$$\text{so } TS = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$$

←
order
right to left

Q5. $R = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$  $y = -x$

or formula $y = (\tan \theta)x$
with $\theta = -\frac{\pi}{4}$

$$S = \begin{pmatrix} 5 & 0 \\ 2 & 5 \end{pmatrix}$$

R then S SO $SR = \begin{pmatrix} 5 & 0 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

$$= \begin{pmatrix} 0 & -5 \\ -5 & -2 \end{pmatrix}$$

Q6. $S = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

x stretch: 4

reflect z

AS S then R

$$RS = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Q7. $y = \frac{1}{\sqrt{3}}x$ let $\tan \theta = \frac{1}{\sqrt{3}}$ $\theta = 30^\circ$

SO $R_1 = \begin{pmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

$y = \sqrt{3}x$ let $\tan \theta = \sqrt{3}$ $\theta = 60^\circ$

SO $R_2 = \begin{pmatrix} \cos 120^\circ & \sin 120^\circ \\ \sin 120^\circ & -\cos 120^\circ \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$

R_1 then R_2

$$R_2 R_1 = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Q8. S then T so $TS = \begin{pmatrix} 0 & 3 \\ -1 & 0 \end{pmatrix}$

Lots of options but seems

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 3 \\ -1 & 0 \end{pmatrix}$$

well $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ by reflect in $y = -x$

then $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 3 \\ -1 & 0 \end{pmatrix}$ by stretch x by -3

$$\text{so } TS = \begin{pmatrix} -3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ -1 & 0 \end{pmatrix}$$

T is stretch x by -3

S is reflect in line $y = -x$

Q9. $R_1 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $R_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $R_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

$x\text{-axis}$ $y\text{-axis}$ $z\text{-axis}$

so $R_3 R_2 R_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$= -I_3$$

Advanced skills

Q10. $R_1 = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

Clockwise by ϕ is Anticlockwise by $360^\circ - \phi$

so $R_2 = \begin{pmatrix} \cos(360 - \phi) & -\sin(360 - \phi) \\ \sin(360 - \phi) & \cos(360 - \phi) \end{pmatrix}$

$$= \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix}$$

R_1 followed by R_2 is $R_2 R_1$

$$\text{SO } R_2 R_1 = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$= \begin{pmatrix} \cos \theta \cos \phi + \sin \theta \sin \phi & \cos \theta \sin \phi - \sin \theta \cos \phi \\ \sin \theta \cos \phi - \cos \theta \sin \phi & \sin \theta \sin \phi + \cos \theta \cos \phi \end{pmatrix}$$

↓ Addition angle formula

$$= \begin{pmatrix} \cos(\theta - \phi) & -\sin(\theta - \phi) \\ \sin(\theta - \phi) & \cos(\theta - \phi) \end{pmatrix}$$

SO Anticlockwise $\theta - \phi$

$$\text{Q11. a) } \cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta}$$

$$\frac{1}{\cos^2 \theta}$$

$$= \frac{1 - \tan^2 \theta}{\sec^2 \theta} = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\sin 2\theta = 2 \sin \theta \cos \theta = \frac{2 \sin \theta \cos \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$

$$= \frac{2 \tan \theta}{\sec^2 \theta} = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

b) As $y = mx$ has $\tan \theta = m$

then $\cos 2\theta = \frac{1-m^2}{1+m^2}$, $\sin 2\theta = \frac{2m}{1+m^2}$

so

$$M = \begin{pmatrix} \frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2} \\ \frac{2m}{1+m^2} & -\frac{1-m^2}{1+m^2} \end{pmatrix} = \frac{1}{1+m^2} \begin{pmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{pmatrix}$$

c) Its true if $M^2 = I_2$ as then $M^3 = MI_2 = M$,
 *& formally by induction*

Now

$$M^2 = \frac{1}{(1+m^2)^2} \begin{pmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{pmatrix} \begin{pmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{pmatrix}$$

$$= \frac{1}{(1+m^2)^2} \begin{pmatrix} (1-m^2)^2 + 2m^2 & 2m(1-m^2) + 2m(m^2-1) \\ 2m(1-m^2) + 2m(m^2-1) & (1-m^2)^2 + 2m^2 \end{pmatrix}$$

$$= \frac{1}{(1+m^2)^2} \begin{pmatrix} (1+m^2)^2 & 0 \\ 0 & (1+m^2)^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Formally for all n , knowing $m^2 = I_2$

True for $n=1,2$, assume true for $n=k$

then

$$M^{k+1} = M^k M = \begin{cases} M M & \text{if } k \text{ even} \\ I_2 M & \text{if } k \text{ odd} \end{cases}$$

$$= \begin{cases} M^2 & \text{if } k \text{ even} \\ M & \text{if } k \text{ odd} \end{cases} = \begin{cases} I_2 & \text{if } k \text{ even} \\ M & \text{if } k \text{ odd} \end{cases}$$

↑

$n = k+1$

So even k

is odd n and
odd k is even n

So true for $k+1$.

Hence if true for k true for $k+1$.

So by induction true for all integers

as true for $n=1$ (and 2)